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RATIONAL DISAGREEMENT

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ABSTRACT

We propose rational disagreement as a formal framework for analyzing seemingly irrational behavior that can persist despite the wide availability of objective information in a steady-state. Agents are rational in that they correctly anticipate the distribution of aggregate outcomes, yet disagree about which specific individuals perform better than others. Notably, the subjective belief of any individual may be objectively correct.

We illustrate the key concepts with a simple entry game. We show how unordered individual outcome distributions can be identified solely from aggregate statistics. We then characterize the resulting game, define its Nash equilibria, and develop a statistical test for the null hypothesis of agreement. Finally, we situate our framework within the broader literature on Bayesian games, behavioral biases, and the rational expectations hypothesis.

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1 Introduction

Empirical studies, along with a wealth of anecdotal evidence, suggest that seemingly irrational behavior persists in economic, financial, and business settings despite overwhelming objective information. A striking illustration is the planning fallacy identified by Kahneman and Tversky (1982), in which individuals systematically underestimate the time and cost required to complete projects, even when they know that projects, in the aggregate, take longer and cost more than planned. Readers will readily recognize this bias in academic settings, where scholars routinely overestimate the number of papers they will write, the speed with which they will complete them, and the quality of the final output.[1] Similar behavior is found in management surveys in entrepreneurship and venture capital (Koellinger et al., 2007; Ben-David et al., 2013), and others confirm that individuals display excessive optimism, despite the fact that they are collectively. In an experimental study by Camerer and Lovallo (1999), most subjects in an entry game predict that their own profit will be positive even though they (correctly) forecast total industry profit to be negative.

The surprising aspect of this behavior is not the presence of optimism per se, but rather its persistence despite the availability of vast amounts of objective data on the time and effort required to complete such tasks. A natural explanation, following Kahneman and Tversky (1982), is to attribute this behavior to incorrect or irrational beliefs—that is, to deviations from what are perceived to be rational benchmarks.

This paper proposes, instead, that rational choice and group interaction can account for these patterns of behavior. We begin with the observation that if past data show that a few projects are completed on time and on budget, while others are not, then each individual may rationally believe that their own project has higher odds of success than others. Since it is impossible to determine ex ante who is right and who is wrong, we refer to this phenomenon as *rational disagreement*. We define rational disagreement as a situation where agents rationally and correctly anticipate the distribution of aggregate outcomes, yet disagree about which specific individuals perform

[1] This fallacy has been extensively documented in the literature; see Buehler et al. (2010) for a survey. Kahneman (2011) goes as far as calling optimism that underlies it, "the engine of capitalism."

better than others.

Formally, consider a cohort of n individuals in a stationary environment, each facing a known objective distribution of aggregate statistics of outcomes, such as productivity levels, project completion times, or success rates in a contest. Knowledge of these aggregates is derived from an (unmodeled) learning process from past observations.

Our first set of results shows that, when outcomes in each cohort are independent, the aggregates can be used to identify a unique unlabeled collection (formally, a *multiset*) of n generating individual distributions. These distributions need not be identically distributed, allowing for within-cohort heterogeneity that is at the heart of our model of disagreement. This identification result holds quite generally and may be of independent interest beyond our work.

Our second main result links identification to individuals' beliefs. While learning forces rational individuals to agree on aggregate information, there remains ample scope for *positional disagreement*, reflecting their uncertainty about their standing within the cohort. As an example, suppose that the objective aggregate distribution implies a generating multiset consisting of one distinguished individual, with the others being merely "average." While all individuals agree on this generating multiset, they need not agree on the identity of the distinguished individual. In the planning fallacy described above, everyone agrees that projects run overtime and over budget on average, except for a few run by talented managers. Then most, or even all, individuals may rationally believe they are the talented ones, since no evidence exists to either confirm or contradict such beliefs.

Our third contribution is to embed this model of disagreement in finite-action games where payoffs depend on players' outcomes as well as their actions. Players hold rational beliefs, in the sense of being consistent with aggregate information, and maximize expected payoffs. We model this as a standard strategic-form game in which players' payoffs reflect their subjective beliefs, and to which the usual equilibrium concepts apply. Since players have different perceptions of the "true" game being played, behavior in a *rational disagreement equilibrium* can be different from that arising in any equilibrium of an objective game.

Finally, we explore the possibility of empirically testing the implications

of disagreement. While players' subjective beliefs are unobservable, their implications on observable outcomes are. Section 4.4 introduces a framework to test for (dis-) agreement even when beliefs and (mixed) strategies are unobservable and vary over time.

⋆ ⋆ ⋆

Rational disagreement has several important implications and connections to the literature. A key distinguishing feature of our approach is its non-paternalistic nature: no outside observer—be it the modeler, econometrician, or benevolent planner—can objectively claim knowledge superior to that of the individuals, even when it is common knowledge that the individuals' beliefs cannot all be correct. This stands in contrast to the common practice in the study of behavioral biases, which treats optimism as an error that an outside observer can discern and, in principle, rectify.

Another key feature of our analysis is that any bias that arises is fundamentally a group phenomenon that disappears in representative-agent settings. The motivating examples introduced above all involve the collective behavior of firms, investors, or entrepreneurs. Our conclusions are consistent with a remarkable behavioral pattern, well-documented in social psychology, that individuals readily identify biases in others while failing to recognize them in themselves (see Section 5.2). Individuals in our model are rational decision makers who regard their own beliefs as carefully reasoned and objectively justified, and therefore attribute the collective biases they observe in their environment to errors in the judgments of others.

Our analysis also connects observed collective biases to objective features of the environment. Individuals whose outcome distributions exhibit little heterogeneity cannot rationally maintain the belief that they are distinguished or special, regardless of how strong their predispositions toward such beliefs may be. Rampant collective biases are therefore more likely to arise in activities with greater heterogeneity—such as entrepreneurship, speculation, the performing arts, and academia—than in activities with more uniform outcomes, such as manual or repetitive tasks.

Finally, rational disagreement is a version of the Rational Expectations Hypothesis, broadly understood, as Lucas (1978) notes in his seminal paper on asset pricing, as the "property likely to be (approximately) possessed

by the outcome of [an] unspecified process of learning and adapting." Our model refines this insight for multi-agent settings: agreement about the aggregates is also reached because individuals act as "macroeconometricians, [who] forecast to the best of their abilities and knowledge, given the available data" (Uhlig, 2012). A similar logic underpins our framework. Our analysis suggests an important qualification: even when individuals agree on aggregates, rational disagreement may persist because individuals' labels cannot be inferred from past data.

The plan for the paper is as follows. Section 2 presents a simple oligopoly entry example that previews our analysis. Section 3 presents the identification results, while Section 4 introduces rational beliefs, strategic interactions, and potential empirical tests of (dis-) agreement. Section 5 explores the implications of our analysis for the existing literature in economics and psychology and the connection with rational expectations. It is there that we discuss the related literature and can do so in light of the framework presented in this paper.

2 Overview and Example

The purpose of this section is to provide an example that showcases our method and the key ideas. The example builds on a familiar entry game between two firms that produce a homogeneous good that sells at a price P. The novel part is what we do with respect to beliefs.[2]

There are two risk-neutral firms, denoted 1 and 2, who face a known demand function $D(P) = 2 - P$. Firms simultaneously decide whether to enter after observing their private entry costs $\kappa_i \in [0, 1]$, where $i = 1, 2$ and $i \neq j$ are always implied to avoid repetition. Upon entry, the entering firms draw marginal cost parameters $c_i \in \{0, 1\}$, then compete in quantities. Thus, a firm is characterized by the pair of parameters $x_i = (\kappa_i, c_i)$, which we refer to as "outcomes" in the rest of the paper.

Firms' payoffs are standard:

1. The payoff of a firm that does not enter is zero.

2. If only Firm i enters, it selects a quantity q_i that maximizes its mo-

[2] Calculations for this example are standard and are collected in the online appendix.

nopolistic profits, gross of entry cost, $\pi(q_i) = q_i(2 - q_i - c)$. Its profit is therefore $\pi = 1$, if $c_i = 0$ and $\pi = 1/4$, if $c_i = 1$. Firm j' profit, in this case, is zero.

3. If both firms enter, they compete as a duopoly and earn profits that depend on their relative cost positions: (a) If $c_1 = c_2 = 0$, then $\pi_1 = \pi_2 = 4/9$; (b) If $c_1 = c_2 = 1$, then $\pi_1 = \pi_2 = 1/9$; and (c) If, $c_i = 0, c_j = 1$, then $\pi_i = 1, \pi_j = 0$.

Firm i's strategy is an entry threshold $a_i \in [0, 1]$, with this firm entering if $\kappa_i \leq a_i$. Its payoff, net of entry cost, is denoted $V_i((a_1, x_1), (a_2, x_2))$.[3]

Firm i has a belief P_i over outcome profiles (x_1, x_2) or $(\kappa_1, c_1, \kappa_2, c_2)$, under which these parameters are independent, and κ_1 and κ_2 are uniformly distributed over the interval $[0, 1]$. A firm enters if its entry cost does not exceed its expected stage-game profit.

The focus of disagreement in this example is the firms' beliefs about marginal costs. Crucially, we require beliefs to be *rational* in the sense of being consistent with aggregate information. This consistency is the hallmark of rational expectations and is motivated by the intuition that firms observe a large number of past instances of similar interactions in a stationary environment. Section 3 formally defines the *aggregate distribution*, denoted $\mathbf{P}$, which the reader should think of as anonymized information that describes the frequencies of the outcomes of interest but not the identities of the firms.

The aggregate distribution in our example is summarized by the probability distribution (f_0, f_1, f_2) of the statistic $c_{sum} = c_1 + c_2$, with f_k indicating the probability of $c_{sum} = k$. This distribution is generated by two individual probabilities $\mu_g \geq \mu_b$, with one firm drawing $c = 0$ with probability μ_g (the "good" firm), and the other with probability μ_b (the "bad" firm) independently. Note that the aggregate information (f_0, f_1, f_2) does not distinguish between $(c_1 = 0, c_2 = 1)$ and $(c_1 = 1, c_2 = 0)$.

In this simple example, it is easy to see how the individual probabilities μ_g and μ_b can be identified from the aggregate distribution: they can be

[3] In the formal discussion of games, we assume actions are finite for simplicity, while strategy spaces in the example are continuous. These can be replaced with appropriately discretized strategy spaces without substantive changes.

deduced from $f_1 = \mu_g \mu_b$ and $f_2 = \mu_g(1 - \mu_b) + (1 - \mu_g)\mu_b$.[4] This simple logic breaks down, however, in the general case with an arbitrary number of individuals and non-binary outcomes. A substantial contribution of this paper is to demonstrate that this insight generalizes, yielding the remarkable result that aggregate information is sufficient to identify a unique *unordered* set (formally, a *multiset*) of individual probabilities for any number of individuals and outcomes in a standard Borel space. This generalization is far from straightforward and requires the deployment of a number of novel tools that may be of independent interest.

While rationality requires firms to agree that there is one good firm and one bad firm, they may disagree on which is which. This is the fundamental feature of our approach: rationality imposes meaningful restrictions on individuals' beliefs about the aggregates, but not on their beliefs about their individual positions given these aggregates.

The two probabilities μ_g, μ_b determine that the state of the game is either the assignment (μ_g, μ_b), where Firm 1 is the good firm and Firm 2 is the bad one, or (μ_b, μ_g), where the roles are reversed. The probability Firm i assigns to (μ_g, μ_b), denoted p_i, determines its beliefs P_i over the outcome pairs (x_1, x_2). We say that the firms are in agreement if $p_1 = p_2$ and are in disagreement otherwise. Facing the entry strategy a_j of its opponent, Firm i enters if its expected stage-game profit is at least as high as its entry cost. The firms play a Nash equilibrium (a_1, a_2) of this entry game. The following cases illustrate the implications of (dis-) agreement:

Symmetric rational agreement: $p_1 = p_2 = 0.5$, so the two firms agree that the assignments (μ_g, μ_b) and (μ_b, μ_g) are equally likely. This is the familiar symmetric benchmark that rules out individual heterogeneity as a source of disagreement.

Asymmetric rational agreement: $p_1 = p_2 > 0.5$, where firms agree that Firm 1 is the likely cost leader with probability p_i.[5]

Symmetric rational disagreement: $p_1 > 0.5$ and $p_2 = 1 - p_1$, where

[4] The corresponding equation for f_2 is implied by the fact that the f_k's add up to 1.

[5] In our numerical calculations of average profit, we assume that these probabilities are correct, *i.e.,* , coincide with the true probabilities that Nature uses to assign roles. In principle, this need not be the case.

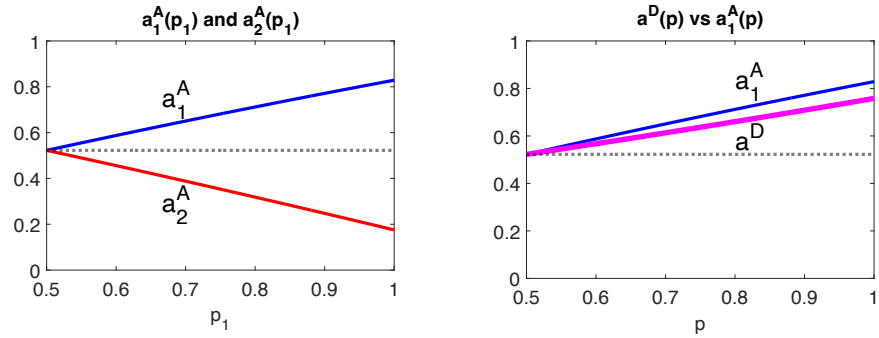

Figure 1: *Suppose* $\mu_g = 0.8$ *and* $\mu_b = 0.2$. *The left panel shows* $a_i^A(p_1)$ *for* $i = 1, 2$ *as a function of* p_1 *in the asymmetric agreement equilibrium. The right panel compares the strategy* $a^D(p)$ *of the symmetric disagreement equilibrium to the agreement strategy* $a_1^A(p)$.

each firm believes it is the likely cost leader: Firm 1 believes the assignment (μ_g, μ_b) is more likely, while Firm 2 (symmetrically) believes the opposite.

In all three cases, beliefs conform with the implications of the aggregate distribution that there is one firm of each cost type μ_g and μ_b. The assignment of these types (either (μ_g, μ_b) or (μ_b, μ_g)), on the other hand, is a latent variable that cannot be identified from aggregate information.

The agreement cases correspond to the classical conception of *rational expectations* where a single probability distribution is "shared by (1) all of the agents within the model, (2) the econometrician estimating the model, and (3) nature, also known as the data-generating mechanism" (Sargent (2008)). These cases illustrate that agreement on the data-generating process is a logical possibility, but requires coordinating expectations on a latent variable for which no data exist. The symmetric disagreement case, by contrast, is radically different since no probability distribution over (μ_g, μ_b) and (μ_b, μ_g) is consistent with the firms' belief profile. No amount of aggregate information, however abundant, can refute a firm's belief that it is the likely cost leader.

Figure 1 illustrates the three cases above. In both panels, the symmetric rational agreement equilibrium is shown as as a reference point with the dotted line at the equilibrium strategies $a_1 = a_2 \approx 0.52$. The equilibrium

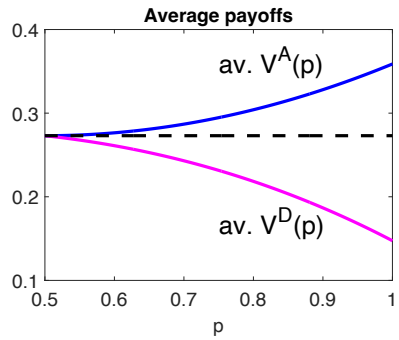

Figure 2: *The Figure compares average total payoffs $\bar{v}^A(p)$ and $\bar{v}^D(p)$, i.e., average total profits net of total average entry costs, for the agreement equilibrium (assuming that the beliefs are correct) and the disagreement equilibrium, at p. The dashed black line indicates the lowest average total payoffs that can arise in any agreement equilibrium, which happens to be the payoff at the symmetric rational agreement equilibrium $p = 0.5$. Average total payoffs in a disagreement equilibrium are always lower, except at $p = 0.5$. This can be used to develop a statistical test, distinguishing agreement play from its alternative.*

entry strategies, $a_1^{\mathrm{A}}(p_1)$, $a_2^{\mathrm{A}}(p_1)$ are shown in the left panel. The two coincide at the symmetric rational agreement case, $p_1 = 0.5$. As p_1 increases, the strategies diverge, with the likely cost leader entering with higher probability.

The symmetric disagreement equilibrium entry strategy $a^{\mathrm{D}}(p)$ is shown in the right panel and is contrasted with the agreement equilibrium strategies $a_1^{\mathrm{A}}(p)$. Symmetric disagreement leads to more aggressive entry strategies. While the game's strategic substitutability tempers entry, the net effect is that both firms behave more aggressively compared to the symmetric rational agreement prediction.

Firms' beliefs are not directly observable and therefore cannot be tested. The implications of these beliefs on entry and realized profits, on the other hand, can be tested against data. In Section 4.4, we provide a theory for testing hypotheses that is valid even when the underlying unobservable beliefs and the equilibrium selection (in games with multiple equilibria) change over time.

As an application of that result, suppose that the entry game has been

played for cohorts $s = 1, \ldots, t$, with independent draws of outcomes and where beliefs can vary across cohorts. Suppose that total firm payoffs $v_s = v_{1,s} + v_{2,s}$, i.e., total profits net of total entry costs, are observable. We wish to test whether they accord with agreement play. For any given $p_1 = p$, expected average total payoffs for rational agreement and rational disagreement equilibria are shown in Figure 2. For any given p, average total rational agreement profits (with correct beliefs) are always above the black dashed threshold at 0.27. Therefore, define

$$F_t = \frac{1}{t} \sum_{s=1}^{t} (0.27 - v_{1,s} - v_{2,s}) = \frac{1}{t} \sum_{s=1}^{t} f(v_{1,s}, v_{2,s})$$

for $f(v_1, v_2) = 0.27 - v_1 - v_2$. Under the conditions stated in section 4.4, $\limsup F_t \leq 0$ almost surely. For example, this is the case if the agreement equilibrium for the same value p_1 is played every period. Therefore, if $F_t \leq 0$ can be statistically rejected, it follows that the agreement equilibrium has not been played for a positive fraction of periods. Average disagreement equilibrium profits can be lower than average agreement equilibrium profits, but they have a lower bound, too. Therefore, a similar test can be developed to possibly reject the hypothesis that a rational agreement equilibrium or a rational disagreement equilibrium has almost always been played.

3 Identification

This section formally defines the concept of "aggregates" and establishes identification results for the underlying individual distributions, which may be of independent interest.

3.1 Model and Notation

Consider a set of independent random variables $x_i,\ i = 1, \ldots, n$, interpreted as the random outcomes of n individuals in a cohort. Each x_i takes values in a standard Borel space X with distribution μ_i.[6] Let $S = X^n$ denote the product space of outcome profiles with its σ-algebra denoted Σ.

[6] To minimize repetition, and unless stated otherwise, we assume throughout the paper that Borel spaces are endowed with their Borel σ-algebras, product spaces with the product σ-algebra, and that all events and functions are measurable.

Write $\Delta(A, \mathcal{A})$ to denote the set of probability distributions on a set A with σ-algebra $\mathcal{A}$, dropping references to $\mathcal{A}$ when it is clear from the context. In the case of the product space S, we distinguish between $\Delta(S)$ and its Borel subset $\Delta_{\text{Indep}}(S)$ consisting of independent probability distributions (see Lemma A.3).

3.1.1 Symmetry: To formalize the idea that labels are arbitrary, recall that an event $E \in \Sigma$ is *symmetric* if for every permutation ϱ of the labels $\{1, 2, \ldots, n\}$:

$$E = \varrho(E),$$

where $\varrho(E)$ is the event $\{(x_{\varrho(1)}, \ldots, x_{\varrho(n)}) : (x_1, \ldots, x_n) \in E\}$. Symmetric events define a measurable space $(S, \mathscr{S})$ where $\mathscr{S} \subset \Sigma$ is the sub-σ-algebra generated by symmetric events. An R^k-valued function $\xi = \xi(x_1, \ldots, x_n)$ is *symmetric* if it is measurable with respect to $\mathscr{S}$.[7] Statistics such as the mean and the variance are examples of symmetric functions.

3.1.2 Multisets: A standard way to represent collections of unordered objects with multiplicities is the concept of a *multiset*, or *bag*, which we distinguish with square brackets and denote with boldface letters. For example, $\boldsymbol{a} = [4, 4, 7]$ as a multiset of numbers consists of three distinct unordered objects and is different from either the set $\{4, 7\}$ or the tuple $(4, 4, 7)$. In the multiset $\boldsymbol{a}$, "4" has multiplicity of 2 while "7" has multiplicity of 1.

Section A.1 details some of the mathematical concepts related to multisets and their representation as elements of a quotient space.

3.1.3 Assignments and the Aggregate Distribution: Given a multiset $\boldsymbol{\mu} = [\mu_1, \ldots, \mu_n]$ in $\Delta(X)$, an *assignment* is a bijection

$$\pi : \{1, \ldots, n\} \to [\mu_1, \ldots, \mu_n]$$

with the interpretation that individual i is assigned distribution $\mu_{\pi(i)}$.[8] Any such assignment uniquely defines the product distribution:

$$P_\pi = \mu_{\pi(1)} \times \cdots \times \mu_{\pi(n)} \in \Delta_{\text{Indep}}(S).$$

[7] That is, $\xi^{-1}(A) \in \mathscr{S}$ for every Borel set in R^k.

[8] Bijections involving multisets need to be defined carefully. See Section A.1, where an equivalent definition in terms of permutations is provided.

The assignment of individuals to distributions is a latent variable that cannot be identified from aggregate data. To state this formally, let:

$$\Phi : \Delta_{\text{Indep}}(S) \to \Delta(S, \mathscr{S})^{9}$$

be the mapping that assigns to every tuple $(\mu_1, \ldots, \mu_n)$ of probability distributions the aggregate distribution $\mathbf{P}$ given by:

$$\mathbf{P}(E) = (\mu_1 \times \cdots \times \mu_n)(E), \text{ for } E \in \mathscr{S}.$$

Clearly, $\mathbf{P}$ depends on the underlying multiset of distributions $[\mu_1, \ldots, \mu_n]$ and not on the particular tuple chosen. We call $\mathbf{P} \in \Delta(S, \mathscr{S})$ in the range of Φ an *aggregate distribution* as it encodes all the properties of the data-generating process that remain invariant to the way the individuals are labeled.

3.2 Identification of the Aggregate Distribution

Consider an environment with an unknown distribution $P \in \Delta_{\text{Indep}}(S)$ and let $\mathbf{P} = \Phi(P)$ be the aggregate distribution it generates. We assume that $\mathbf{P}$ is known through an unmodeled steady-state learning process based on observations of past cohorts. The identification problem is to recover P from the aggregate distribution $\mathbf{P}$. Since aggregate data suppresses all information about labels, we can at most recover the multiset of generating distributions, and hence identify P only up to permutations of these labels.

Formally, we say that $\Delta_{\text{Indep}}(S)$ is *identified (up to permutations of labels)* if, for every $\mathbf{P}$ in the range of Φ, $\Phi^{-1}(\mathbf{P})$ consists of distributions that share the same multiset of marginals.

The remainder of this section introduces three identification theorems in increasing generality. We begin with the case of a binary outcome space ($X = \{0, 1\}$) which we single out for its simplicity, its intuitive proof, and the light it sheds on the problem's structure. In this case, x_i is a Bernoulli trial, where the probability of $x_i = 1$ is p_i, and the aggregate distribution is the Poisson Binomial.

Theorem 1. *Suppose that $X = \{0, 1\}$. Then the set of Poisson Binomial distributions is identified up to permutations of the labels.*

[9] As shown in Lemma A.4, Φ is a measurable mapping between standard Borel Spaces.

The proof of this theorem, like all other proofs in this paper, appears in the Appendix.

In the Bernoulli case, the given aggregate distribution $\mathbf{P}$ is the Poisson Binomial and characterized by the n probabilities $f_k = \mathbf{P}(S = k)$ of observing $k = 1, \ldots, n$ realizations of 1 ($k = 0$ is excluded since probabilities add up to 1). From this information, the goal is to determine the multiset of parameters $[p_1, \ldots, p_n]$ of the underlying Poisson Binomial. If the p_i's happen to be equal, the aggregate distribution is a Binomial distribution whose single parameter can be readily computed. The challenging case is when the p_i's are not necessarily equal.

The proof is based on factoring the probability-generating function of the symmetric function $\xi = x_1 + \cdots + x_n$. Independence ensures that we can write this function as a polynomial of degree n with coefficients determined by the f_k's. The Fundamental Theorem of Algebra guarantees that this polynomial has a unique multiset $[z_1, \ldots, z_n]$ of n complex roots. We show that these roots are real and can be used to recover the underlying Bernoulli probabilities. These probabilities have the same multiplicities as the roots of the polynomial (Equation (5) in the proof). In the special case where the data-generating process is Binomial, the identified multiset is $[p, \ldots, p]$, and p is the parameter of the Binomial distribution.

Next, we turn to the case where outcomes are in an arbitrary finite set. The goal now is to identify the multiset $[p_1, \ldots, p_n]$ of parameters of the underlying Poisson Multinomial distribution, where each p_i is a probability distribution on the finite set of outcomes. The following theorem establishes identification for these distributions and is a key step in proving the subsequent general identification theorem.

Theorem 2. *Suppose that $X = \{x_1, \ldots, x_k\}$ is an arbitrary finite set. Then the set of Poisson Multinomial distributions is identified up to permutations of the labels.*

The argument used in the binary case cannot be carried over to this setting, for reasons detailed in the Appendix. To handle the multinomial case in full generality, we rely instead on subtle arguments involving multivalued probability-generating functions and the fact that the ring of polynomials with real coefficients is a Unique Factorization Domain.

We finally turn to the general case where outcomes are in a standard Borel space. The set of probability distributions in this case is the set of all products of Borel measures on the n coordinates.

Theorem 3. *Suppose that X is a standard Borel Space. Then the set $\Delta_{\text{Indep}}(S)$ is identified, up to permutations of the labels.*

The proof builds on Theorem 2 since, as discussed in the Appendix, identification in the binary case is not sufficient.

4 Rational Disagreement Equilibrium

4.1 Actions and Payoffs

Consider a game with n players, arbitrarily labeled $1, \ldots, n$, each taking an action $a_i \in A_i$ in a finite set A. Payoffs are given by the function:

$$V : A^n \times S \to R^n$$

where $x_i \in X_i = X$ is a random outcome in a standard Borel space X. Mixed strategies are represented by $\Delta(A_i)$ and denoted by α_i. We require that:

Assumption 1.

1. *V is measurable; and*
2. *V is equivariant: for any permutation ϱ of the individuals' labels $\{1, \ldots, n\}$, we have:*

$$V\big((a_{\varrho(1)}, y_{\varrho(1)}), \ldots, (a_{\varrho(n)}, y_{\varrho(n)})\big) = \big(V_{\varrho(1)}, \ldots, V_{\varrho(n)}\big)\,.$$

Equivariance ensures that the players' arbitrary labeling does not matter. In light of this, we fix arbitrary labels for the rest of this section.

Each individual has a belief P_i over outcome profiles S. Given such P_i, define player i's *reduced-form* payoff function

$$\mathcal{V}_i : A^n \to R$$

by integrating with respect to P_i:

$$\mathcal{V}_i(a_1, \ldots, a_n; P_i) = \int_S V_i(a_1, \ldots, a_n; x_1, \ldots, x_n)\, dP_i(x_1, \ldots, x_n).$$

Payoffs are extended to mixed strategies in the usual way.

Any belief profile $(P_1, \ldots, P_n)$ defines the *(reduced-form) game*:

$$G(P_1, \ldots, P_n; A, V)$$

where, for each player i, A_i is the common finite set of actions A and $\mathcal{V}_i(\,\cdot\,; P_i)$ is the (reduced-form) payoff function determined by this player's belief. For the remainder of the paper, we suppress references to the action set A and the payoff function V, which we fix throughout.

Each game $G(P_1, \ldots, P_n)$ is a standard finite-action strategic form game, where equilibrium concepts and their existence can be introduced in the usual way. In particular, a profile of strategies $(\bar{\alpha}_1, \ldots, \bar{\alpha}_n)$ is a Nash equilibrium if for every player i

$$\bar{\alpha}_i \in \arg\max_{\alpha_i} \mathcal{V}_i(\alpha_i, \bar{\alpha}_{-i}\cdot\,; P_i). \tag{1}$$

4.2 Rational Beliefs

The identification results of the last section are statistical in nature. What we are interested in are the beliefs of rational players who observe an aggregate distribution $\mathbf{P} \in \Delta(S, \mathscr{S})$. We begin with a definition:

Definition 1. *A player's belief $P \in \Delta(S)$ is* rational at $\mathbf{P}$ *if $\Phi(P) = \mathbf{P}$.*

That is, the probabilities of the symmetric events under a rational belief P and the aggregate distribution $\mathbf{P}$ coincide. The next theorem characterizes rational beliefs. To set the stage, we first note that $\Delta\big(S, \mathscr{S}\big)$ is a standard Borel space, with σ-algebra $\mathscr{G}$.[10] Consider an ex ante stage where a Bayesian decision maker has a prior belief:

$$\tilde{P} \in \Delta\left(\Delta_{\text{Indep}}(S) \times \Delta(S, \mathscr{S})\right)$$

[10] The Borel space $\big(S, \mathscr{S}\big)$ is not standard since $\mathscr{S}$ does not separate points. However, the set of probability distributions on this space is standard Borel. See Lemma A.2 in the Appendix.

representing his uncertainty about pairs $(P, \mathbf{P})$, with marginal distribution on $\Delta\big(S, \mathscr{S}\big)$ denoted ν. Rationality requires consistency with Φ:

$$\tilde{P}\left(\{(P, \mathbf{P}) : \Phi(P) = \mathbf{P}\}\right) = 1.$$

In other words, $\tilde{P}$ puts unit mass on the graph of Φ.

Theorem 4. *Suppose that the players' ex ante beliefs $\tilde{P}_1, \ldots, \tilde{P}_n$ are mutually absolutely continuous. Then, there is a Borel set $A \subset \Delta\big(S, \mathscr{S}\big)$, with $\nu_i(A) = 1$, for $i = 1, \ldots, n$, such that for every $\mathbf{P} \in A$, the posterior beliefs $P_i = \tilde{P}_i(\cdot \,|\, \mathbf{P})$ have the form:*

$$P_i = \sum_{\pi} \big(\mu_{\pi(1)} \times \cdots \times \mu_{\pi(n)}\big) \; w_i(\pi),$$

for a multiset $\boldsymbol{\mu} = [\mu_1, \ldots, \mu_n]$ and non-negative weights satisfying $\sum_\pi w_i(\pi) = 1$, where the sum is taken over all assignments $\pi : \{1, \ldots, n\} \to [\mu_1, \ldots, \mu_n]$.

Let Π be the set of all assignments with values in $\boldsymbol{\mu}$. Consider a random assignment $\tilde{\pi}$ and, abusing notation, use $\tilde{\pi}$ to also denote its distribution $\tilde{\pi} \in \Delta(\Pi)$. The theorem states that a rational belief for player i must have the form $P_{\tilde{\pi}_i}$ for some random assignment $\tilde{\pi}_i$. Such beliefs are forced to agree on the common multiset $\boldsymbol{\mu}$, so the only remaining disagreement must be positional, *i.e.,* confined (almost surely) to differing subjective opinions about the latent variable π.

4.3 Equilibrium

In light of Theorems 4, for the remainder of this paper, we fix a generic aggregate distribution $\mathbf{P}$ with generating multiset $\boldsymbol{\mu}$ and, when convenient, write rational belief profiles explicitly as $(P_{\tilde{\pi}_1}, \ldots, P_{\tilde{\pi}_n})$. The dependence of beliefs on $\mathbf{P}$ and $\boldsymbol{\mu}$ is implicit to simplify the notation.

Definition 2. *A game $G(P_1, \ldots, P_n)$ is an* agreement game *if there is a random assignment $\tilde{\pi}$ such that $P_i = P_{\tilde{\pi}}$ for all i, otherwise it is a* disagreement game*. A profile of mixed strategies is an* agreement (Nash) equilibrium *if it is an equilibrium for some agreement game.*

We interpret an agreement game as one in which players share the same model of the game being played. These are the games readers are familiar

with, and exemplified by the rational agreement setting in the entry game example of Section 2. In such games, payoffs are generated by an objective distribution $P_{\tilde{\pi}}$ that all players agree on.

A disagreement game, by contrast, is one where players may have different perceptions of the game, reflecting their theories $\tilde{\pi}_i$ of how the individual distributions are assigned. The result is subjective games and equilibria, as defined by Kalai and Lehrer (1995), and discussed at length in Section 5.4 below. Formally, write

$$\mathcal{N}(P_1, \ldots, P_n) \subset (\Delta(A))^n$$

for the set of Nash equilibrium distributions of $G(P_1, \ldots, P_n)$.

Definition 3. *A* rational disagreement equilibrium *for* $G(P_1, \ldots, P_n)$ *is an equilibrium* α *such that:*

$$\alpha \notin \bigcup_{\tilde{\pi} \in \Delta(\Pi)} \mathcal{N}(P_{\tilde{\pi}}, \ldots, P_{\tilde{\pi}}).$$

That is, the distribution of actions implied by α cannot be rationalized by any agreement game. Concrete examples of (dis-) agreement games and equilibria appear in Section 2 where the multiset of individual distributions is $[\mu_a, \mu_b]$ and the two possible assignments are (μ_a, μ_b) and (μ_b, μ_a). In the notation of that section, Firm 1 assigns probability p_1 to $\tilde{\pi}_1 = (\mu_a, \mu_b)$, while Firm 2 assigns probability p_2 to $\tilde{\pi}_2 = (\mu_a, \mu_b)$. In the symmetric disagreement case, $p_1 > 0.5$ while $p_2 = 1 - p_1 < 0.5$, indicating that both firms believe they are the likely cost leader. The resulting game, $G(P_{\tilde{\pi}_1}, P_{\tilde{\pi}_2})$, has payoffs that cannot be produced by any common assignment $\tilde{\pi}$. As shown in Figure 1, the unique equilibrium of this game is not an equilibrium of a game in which Nature draws the assignment according to a commonly agreed upon probability. When we introduce players' beliefs about other players' beliefs in Section 5, neither player believes $G(P_{\tilde{\pi}_1}, P_{\tilde{\pi}_2})$ is the true game.

4.4 Empirical Tests of Rational Disagreement

An important question is whether a theory of rational disagreement can be tested against data. Since beliefs are not directly observed, a test can

only rely on the observed implications of these beliefs. This section provides a framework for addressing this question.

Suppose we have access to infinitely many cohorts, indexed by $t = 1, 2, \ldots$, each with the same action set A and payoff function V. The outcomes are stationary and generated by an aggregate distribution $\mathbf{P}$. In any cohort, the game is $G(P_1^t, \ldots, P_n^t)$, where $P_1^t, \ldots, P_n^t$ is a profile of (not necessarily rational) beliefs. Players in cohort t play the unobservable mixed strategy profile α^t. Beliefs and strategy profiles can vary across cohorts.

Define the *state of the game* at a given date as:

$$Z = \big(\Delta(S) \times \Delta(A) \times A \times X \times R\big)^n$$

with a generic element $\{(P_i, \alpha_i, a_i, x_i, v_i)\}_{i=1}^n$ representing the profile of beliefs, mixed strategies, actions, outcomes, and realized payoffs of the players. Assume that an infinite sequence $\mathcal{Z} = \{z^1, z^2, \ldots\} \in Z^\infty$ of states is observed. We require:

Assumption 2. *The sequence $\mathcal{Z}$ is generated by some (unknown) stochastic law Q satisfying:*

1. *The draws z^t are independent over time.*
2. *For every i and t,*
   - *The distribution of the multisets of outcomes $[x_1^t, \ldots, x_n^t]$ is a given aggregate distribution $\mathbf{P}$;*
   - *$(\alpha_1^t, \ldots, \alpha_n^t)$ is a Nash equilibrium for the game $G(P_1^t, \ldots, P_n^t)$;*
   - *a_i^t is drawn from α_i^t;*
   - *$v_i^t = V_i(a_1^t \ldots, a_n^t, x_1^t, \ldots, x_n^t)$.*

The second part of the assumption gathers consistency conditions. The independence requirement is more substantive, yet it allows for dependence between x_i^t and α_i^t. This arises, for example, in asymmetric agreement equilibria, in which individuals are correct about their beliefs P_i^t concerning the distribution of x_i^t. The assumption also allows for nonstochastic intertemporal dependence of the pick of the Nash equilibria α, such as, say, convergence to a particular Nash equilibrium. This allows beliefs and equilibrium selection to vary across cohorts.

A *statistic* is any function

$$f : Z \to R$$

that (1) symmetric, and (2) does not depend on the (unobservable) players' beliefs $\Delta(S)$ or mixed strategies $\Delta(A)$. Symmetry requires that f is invariant to the way the players are labeled.

A statistic measures a property of interest. For example, if v_i^t denotes the realized profit of firm i at time t, then the statistic $\sum_{i=1}^n v_i^t$ reports the realized aggregate industry profit in a given cohort at date t. This is the statistic used in the entry game example in Section 2.

To test the prevalence/absence of disagreement, let

$$\mathcal{A} = \bigcup_{\tilde{\pi} \in \Delta(\Pi)} \mathcal{N}(P_{\tilde{\pi}}, \ldots, P_{\tilde{\pi}}).$$

to be the set of agreement equilibria. Define $\mathcal{D} \subset (\Delta(A))^n$ similarly to denote the set of disagreement equilibria. We require:

Assumption 3. *The statistic f and the data-generating process Q satisfy:*

1. *There exists $M > 0$ such that, for all t,*

$$\mid f(z^t) \mid \leq M, \; Q - a.s.$$

2. $E[f(z^t) \mid (\alpha_1^t, \ldots, \alpha_1^t) \in \mathcal{A}] \leq 0.$

Formulate the hypothesis

$$\mathbf{H}_0 : \text{ For all } t, (\alpha_1^t, \ldots, \alpha_1^t) \in \mathcal{A}, \; Q - \text{a.s.}$$

that the realizations are generated as if players were rational and played an equilibrium of an agreement game $G(P_1^t, \ldots, P_n^t)$ Q-a.s. at all times. Define

$$F_t = \frac{1}{t} \sum_{s=1}^{t} f(z^s)$$

to be the date-t sample average of the statistic f. We are interested in the fraction of times that an agreement equilibrium was played:

$$\Psi_t = \frac{1}{t} \sum_{s=1}^{t} 1_{(\alpha_1^s, \ldots, \alpha_n^s) \in \mathcal{A}}$$

which is not directly observable.

Theorem 5. *Under assumptions 2 and 3, for Q-a.e. sequence of observations $\mathcal{Z}$,*

1. $\limsup_{t\to\infty} F_t \leq 0$ *under* $\mathbf{H}_0$*; and*
2. $\liminf_{t\to\infty} F_t > 0$ *implies* $\limsup_{t\to\infty} \Psi_t < 1$*,*

5 Rationality, Disagreement, and Collective Biases

The framework of reduced-form games introduced in the last section specifies beliefs only insofar as is necessary to determine equilibrium strategies. While sufficient for describing equilibrium behavior, this framework is too limited to capture social and economic interactions in which individuals' perceptions of their opponents' beliefs and biases play a central role. Modeling these phenomena requires a *Theory of Mind*—a representation of how players recognize that their opponents hold distinct beliefs, and how these beliefs can be used to predict, interpret, and respond to their behavior. This section develops such a theory and connects it to the literature in game theory, social psychology, and rational expectations.

5.1 A Theory of Mind

Bayesian games provide a natural way to enrich reduced-form games with a Theory of Mind.[11] The next theorem provides a canonical way to do so:

Theorem 6. *To every reduced-form game $G = G(P_1, \ldots, P_n)$ corresponds a unique Bayesian game with* common knowledge of biases $B = B(P_1, \ldots, P_n)$ *defined by the requirement that the belief profile $P_1, \ldots, P_n$ is common knowledge. The games G and B have the same set of equilibrium distributions $\mathcal{N}(P_1, \ldots, P_n)$.*

Applied to rational beliefs, the states of nature are the assignments Π, so the players' (first-order) beliefs have the form $P_{\tilde{\pi}_i}$, where π_i is player i's

[11] An additional foundational reason to expand the description of beliefs is the idea that Bayesian decision makers should form beliefs about all the uncertainties in their environment—including the beliefs of others.

subjective assignment. Higher-order beliefs are determined by the common knowledge assumption.

We think of the game with common knowledge of biases as a useful device to interpret some of the more puzzling aspects of the analysis. If the beliefs $P_{\tilde{\pi}_1}, \ldots, P_{\tilde{\pi}_n}$ are all different, then in the Bayesian game, everyone agrees that $G = G(P_{\tilde{\pi}_1}, \ldots, P_{\tilde{\pi}_n})$ is not the game being played. In fact, it is common knowledge that each player i is certain that the true game is $G(P_{\tilde{\pi}_i}, \ldots, P_{\tilde{\pi}_i})$ and that his opponents' strategy choices are best responses to the "wrong" beliefs. The game G conveniently assembles pieces of the players' perceptions into a structure that is easy to analyze.

Equilibrium behavior arises, as usual, from the implicit assumption that a fictitious mediator proposes strategies, from which players have no incentive to deviate in light of the strategies of others. An alternative perspective is to view each firm's behavior as the solution to a Bayesian decision problem, without recourse to a mediator. In a disagreement game with a unique Nash equilibrium, the common knowledge of beliefs suffices for any individual agent to deduce the strategy to be chosen by all others and to play best response against these deduced strategies so as to arrive at that same Nash equilibrium.

Contrast this with an alternative we refer to as *stubborn disagreement*. Firms have the same first-order beliefs as above, but higher-order beliefs differ: each firm i believes that firm j shares its own beliefs P_i, when, in fact, it has beliefs P_j. As a result, each firm "stubbornly" believes it is in the asymmetric agreement game, in which it is the cost leader. These believes lead to the strategy profile $\alpha^{SD} = (\alpha_1^A, \alpha_1^A)$, in which both firms pursue the aggressive strategy of the cost leader in an agreement game, and which is more aggressive than the disagreement equilibrium play (α_1^D, α_2^D), as illustrated in the right panel of Figure 1. Put differently, in a disagreement equilibrium, firms correctly deduce the equilibrium play of the other firm, though attributing it to wrong beliefs, while firms wrongly deduce the strategy of the other firm to be the cost follower strategy α_2^A under stubborn disagreement, and are then surprised by the choice of their opponent. Of course, their wrong second-order beliefs are immaterial in a reduced-form Nash equilibrium, where a fictitious mediator supplies them with the strategy of others: they now simply find the strategic choice of their opponent

inexplicable.

5.2 The Psychology of Collective Biases

An important phenomenon in social psychology is the *bias blind spots*, defined as the "broad and pervasive tendency to see [...] the impact of bias on others' judgments while denying its influence on their own" (Pronin et al., 2004). The tendency to readily perceive "biases" in others but not in oneself "generalizes across social, cognitive, and behavioral biases; begins in childhood; and appears across cultures. People show a bias blind spot in high-stakes contexts, including investing, medicine, human resources, and law" (Pronin and Hazel, 2023).[12]

In our model, this phenomenon corresponds to *collective biases*, where members of the cohort agree both that everyone is rational and that the cohort as a whole is not. Since individuals are Bayesian decision makers, they cannot, by definition, perceive biases in themselves. On the other hand, these individuals see biases all around them—even in a stationary environment where everyone knows the objective aggregate distribution.

Kahneman and Tversky (1982) offer a deeper insight into the psychological mechanisms underlying this behavior by distinguishing between *distributional information*—defined as "knowledge about the distribution of outcomes in similar situations"—and *singular information*, which concerns "the particular case under consideration." For example, an entrepreneur's distributional information includes objective, readily accessible facts such as average rates of return on capital and the likelihood of securing venture-capital funding. Singular information, by contrast, encompasses the entrepreneur's subjective assessment of his managerial ability and narratives about past triumphs and failures. According to Kahneman and Tversky, individuals tend to place excessive weight on singular information "even when it is scanty and unreliable, and give insufficient weight to distributional information." The planning fallacy reflects this imbalance: its "signature [...] is not that planners are optimistic, but that they maintain their optimism about the current project in the face of historical evidence to the contrary" (Buehler et al., 2010).

[12] See Pronin (2008) for a comprehensive review.

Without a formal model, it is difficult to define singular vs. distributional information, let alone specify what it means to over-rely on one or the other. In our model, distributional information corresponds to the aggregate distribution $\mathbf{P}$, while singular information is summarized by the individual's subjective belief P_i. Disagreement arises because individuals view their draws as non-repeatable singular events for which no relevant past data exist, and thus see no compelling reason to believe that they are "just another statistic."

A natural explanation of this behavior is to ascribe it, as Kahneman and Tversky (1982) do, to incorrect or irrational beliefs. Such explanations face several well-known challenges. First, how can such biases persist in the face of learning and competition when commonly agreed-upon, objective benchmarks are readily available? Second, explanations based on mistaken beliefs beg the question of what constrains the degree to which individuals who disregard widely available objective information indulge in irrational beliefs. In the case of the planning fallacy, over-optimistic beliefs may range from delusional to moderate or negligible, with no discipline imposed by the objective environment in which individuals find themselves. Our analysis imposes structure on rational beliefs that overcome some of these difficulties.

Collective biases contrast with behavioral explanations in two other respects. First, the view that decisions are made by irrational individuals whose beliefs contradict objective evidence naturally invites calls for corrective interventions. The case for such interventions is less compelling under collective biases since no party can claim knowledge superior to that of the individuals themselves. Second, behavioral explanations emphasize single-agent decision-making while collective biases are fundamentally a group phenomenon that cannot arise in a representative-agent setting ($n = 1$). This view aligns more closely with the common intuition in the literature that, at the individual level, learning eliminates biases in a steady state.

The question of the nature of "biased" beliefs is both fascinating and controversial. Collective biases represent departures from the common prior assumption that dominates much of economic research. At the same time, there is substantial empirical evidence that such biases are central in social, economic, and financial contexts. Another strand of research emphasizes evolutionary pressures as a source of biases and disagreement (*e.g.,* Johnson

and Fowler (2011)). In this paper, we remain agnostic about the origins of beliefs and focus instead on their implications for observed choices. Research in psychology and related fields is, therefore, complementary and invaluable for understanding the origins and manifestations of these biases.

5.3 Consistency and Rational Expectations

Our analysis should be contrasted with the Rational Expectations Hypothesis (REH), which serves as the foundation for many economic models. In its standard form, REH asserts that there is a single probability distribution "shared by (1) all of the agents within the model, (2) the econometrician estimating the model, and (3) nature, also known as the data-generating mechanism" (Sargent (2008)). A common justification of this methodology is that beliefs merge, through an implicit learning process, towards agreement on the true distribution.

Our analysis shares this view with respect to the aggregate distribution $\mathbf{P}$. As argued above, however, agreement on the aggregates falls short of completely pinning down beliefs, leaving room for disagreement that can persist even when individuals learn from shared steady-state data. There are exceptions, of course. When cohort behavior is reduced to a single representative agent, or when all distributions in the multiset are equal, beliefs necessarily conform to the strong form of the REH above. Alternatively, REH can be imposed by fiat—for example, by simply assuming agreement.

These narrow exceptions reinforce our main point, namely that disagreement is the unsurprising—indeed, the natural—state of beliefs. The present paper thus suggests a less sweeping version of the REH: individuals are constrained by Lucas's "process of learning and adapting," but this consistency falls short of enforcing complete agreement.

Another tenet of the strong form of the REH is that "agents' expectations disappear as objects to be specified by the model-builder or to be estimated by the econometrician" (Sargent, 2008). In our analysis, subjective beliefs do not generally disappear as free variables in the model, leaving ample room for social, psychological, and evolutionary forces to shape beliefs in ways that REH simply preclude.

5.4 Additional Literature

Next, we turn to important works in the literature not covered earlier. The definition of rational disagreement equilibrium is similar to Kalai and Lehrer (1995) concept of *subjective equilibrium*. In their framework, players are Bayesian decision makers who best respond to their opponents' strategy choices but may hold different perceptions of the game being played. As always, the challenge with introducing subjectivity is that "anything goes." While initially unconstrained, subjectivity in Kalai and Lehrer (1995) eventually disappears as a result of learning from repeated play. Our approach is related but distinct: players' subjective perceptions of the game are constrained by the fact that they derive from the same objective aggregate distribution.

In a landmark paper, Fudenberg and Levine (1993) introduce a model of steady-state learning in repeated extensive form games. In their model, players are assigned specific roles in the stage game and are repeatedly matched at random with other players in the opponents' roles. The present paper emphasizes individual heterogeneity. The analysis can be extended straightforwardly to environments with observable traits that categorize individuals by role, gender, access to resources, or other characteristics. Each category would then have its own aggregate distribution, and disagreement would arise because of heterogeneity within categories rather than across the entire population.

Benoît and Dubra (2011) consider players' ordinal self-rankings (*e.g.,* how many individuals place themselves in the top decile) in a standard Bayesian setting where individuals receive private signals. They prove an ingenious result characterizing when a signal structure can be found so that Bayesian updating leads to rank optimism. Their characterization imposes bounds on rank optimism, so "a rational population can appear to be twice as confident as reality would suggest, but no more." On the other hand, "data in which 1/2 of the population places itself in the top two deciles cannot be explained as rational" with respect to any signal structure. Disagreement in our setting arises for different reasons and displays different features. First, no private signals are necessary for collective biases to arise. —even though they may be included and need not be limited to ordinal comparisons. An example is overconfidence, where individuals may have

the same estimate of their mean outcome but believe they are exposed to lower variance than others. Disagreement in our setting is constrained by the objective generating multiset, but is otherwise unconstrained.

Finally, this paper focused on overoptimism because it is the empirically relevant phenomenon in our motivating examples. It should be clear that nothing in the theory requires or favors optimism. Individual "biases" arising from Theorem 4 can reflect a population of optimists, pessimists, as well as other interesting belief patterns.

5.5 Conclusions

We have proposed *rational disagreement* as a novel approach for analyzing collective biases that enriches the existing literature. In our formal approach, agents are *rational* in the sense that they correctly anticipate the distribution of aggregate outcomes, yet are in *disagreement* regarding which specific individuals perform better than others. As a result, any individual belief may be correct a priori. An outside observer cannot correct "mistakes" by the participants that would lead to a Pareto improvement, in contrast to much of the behavioural literature.

More precisely, suppose that agents differ in the probability of drawing more favorable outcomes, generating an aggregate distribution of outcomes. We show how the unordered individual outcome distributions can be identified solely from aggregate statistics. To do so, we deploy tools of multivalued probability generating functions and descriptive set theory, among others. The beliefs of agents are beliefs about the random assignment of the individual outcome distributions to all players. We formalize and analyze the resulting games and develop a framework for testing the null hypothesis of agreement. Finally, we situate our framework within the broader literatures on Bayesian games, psychological bias, and the rational expectations hypothesis.

Our framework is suitable for applications in industrial organization, finance, and other fields, and we have laid the groundwork for doing so.

A Appendix

A.1 Multisets

Fix an arbitrary positive integer n, a set A, and its n-fold product A^n. The group $G = G^n$ of permutations $\varrho : \{1, \ldots, n\} \to \{1, \ldots, n\}$ defines an equivalence relation $\sim$ on A^n whose equivalence classes, or *orbits*, are subsets that are closed under the action of G. A multiset $\boldsymbol{a} = [a_1, \ldots, a_n]$ in A is an orbit of A^n under G.

Multisets have two other convenient and equivalent representations. First, the multiset $\boldsymbol{a}$ can be written as:

$$\left[a_1^{(m_1)}, \ldots, a_K^{(m_K)}\right], \tag{$*$}$$

where $a_1, \ldots, a_K$ are distinct elements of A and the m_k's are their *multiplicities*. For example, $\left[a^{(n)}\right] = [a, \ldots, a]$ is a multiset of n distinct objects, all of which are equal as members of A. A second useful representation consists of "flattening" (*) into the set

$$\{(a_k, r) : k = 1, \ldots K, r = 1, \ldots, m_k\}, \tag{$**$}$$

of n distinct elements. A *stabilizer* is any bijection ρ from (**) into itself such that, for every a_k, $\rho(a_k, r) = \rho(a_k, r')$ for some $r, r' \in \{1, \ldots, m_k\}$. A stabilizer reshuffles around the indices r without affecting the distinct elements of the multiset or their multiplicities.

The representation (**) is useful in defining functions between multisets. For example, the function $\eta : [a, a] \to [a, b]$ is a shorthand for $\eta : \{(a, 1), (a, 2)\} \to \{(a, 1), (b, 1))\}$ and the notation $\eta(a)$ refers to either $\eta(a, 1)$ or $\eta(a, 2)$, which is usually clear from the context.

An assignments is a bijection $\pi : \{1, \ldots, n\} \to [a_1, \ldots, a_n]$ that selects the unique an n-tuple $(a_{\pi(1)}, \ldots, a_{\pi(n)})$ from $[a_1, \ldots, a_n]$. Using the representations of multisets (**), an assignment π can be expressed as the equivalence class of bijections:

$$\hat{\pi} : \{1, \ldots, n\} \to \{(a_k, r) : k = 1, \ldots K, r = 1, \ldots, m_k\},$$

where $\hat{\pi}$ is equivalent to $\hat{\pi}'$ if $\hat{\pi}' = \rho \circ \hat{\pi}$ for some stabilizer ρ.

Next, we add measurable structures and probability measures on multisets in X. Let $S_{/G}$ be the quotient space of S with respect to the equivalence relation G, and let $\mathscr{M}$ be the induced quotient σ-algebra. Thus, $(S_{/G}, \mathscr{M})$ is the measurable space of multisets of outcomes $[x_1, \ldots, x_n]$ which, by Lemma A.1, is a standard Borel space.

Although we can define aggregate distributions as elements of $\Delta(S_{/G}, \mathscr{M})$, we find it more convenient to define them on the space $(S, \mathscr{S})$ of outcome profiles with the σ-algebra of symmetric events. Lemma A.2 shows that there is a natural Borel isomorphism between $\Delta(S_{/G}, \mathscr{M})$ and $\Delta(S, \mathscr{S})$. This makes it possible to formally define "$[\mu_1, \ldots, \mu_n]$ is the *generating multiset* of the random variables $x_1, \ldots, x_n$," to mean that the distribution of the random object $[x_1, \ldots, x_n]$ is given by $\mathbf{P} \in \Delta(S, \mathscr{S})$.

A.2 Identification Theorems

Proof of Theorem 1: From the definition of the Poisson Binomial distribution, we have, for every $k = 0, 1, \ldots, n$,

$$f_k = \sum_{\substack{A \subseteq \{1,2,\ldots,n\} \\ |A|=k}} \left(\prod_{i \in A} p_i \right) \left(\prod_{j \notin A} (1 - p_j) \right). \tag{2}$$

Since ξ is a discrete random variable with non-negative values, its probability-generating function is well-defined and given by:

$$\begin{aligned} G(z) &\equiv E[z^{\xi}] \\ &= E[z^{x_1} \times \cdots \times z^{x_n}] \\ &= \prod_{i=1}^{n} E[z^{x_i}] \qquad (3) \\ &= \prod_{i=1}^{n} \left((1 - p_i) + p_i z \right), \qquad (4) \end{aligned}$$

where (3) follows from independence, and (4) is the product of the probability-generating functions for Bernoulli random variables. Expanding (4) and

sorting the powers of z, we have:

$$\begin{aligned} G(z) &= \sum_{k=0,1,\ldots,n} \sum_{\substack{A\subseteq\{1,2,\ldots,n\}\\ |A|=k}} \left(\prod_{i\in A} p_i\right)\left(\prod_{j\notin A}(1-p_j)\right) z^k \\ &= \sum_{k=0,1,\ldots,n} f_k z^k, \end{aligned} \tag{5}$$

where (5) is obtained by substituting in (2).

Claim. *Let $l \leq n$ be the largest integer such that $f_l > 0$. Then precisely $n-l$ of the parameters are 0.*

Proof of the Claim: The claim is obvious when $f_n > 0$. If $f_n = 0$ then, by inspecting (2), $p_i = 0$ for at least one i which we set, without loss of generality, to $i = n$. If $l < n-1$ and thus $f_{n-1} = 0$, note that all its summands in (2) are non-negative and thus are all equal to zero. In particular, $(1-p_n)\prod_{i=1}^{n-1} p_i = \prod_{i=1}^{n-1} p_i = 0$. Thus, another p_i, say, p_{n-1} is equal to 0. Continuing this argument recursively delivers the claim. ■

Assume, first, that $f_n > 0$. From the claim, it follows that the parameters of the Poisson Binomial distribution are all strictly positive. Then equation (5) defines a polynomial of degree n with coefficients $f_0, f_1, \ldots, f_n$. By the Fundamental Theorem of Algebra, this polynomial has a unique multiset of complex roots $[z_1, \ldots, z_n]$. This yields the identity

$$\prod_{i=1}^{n}((1-p_i)+p_i z) = \sum_{k=0,1,\ldots,n} f_k z^k \tag{6}$$

which implies that the multiset of roots $[z_1, \ldots, z_n]$ coincides with the solutions of the n degree 1 polynomials on the LHS. Thus, each root z_i must satisfy the equation $(1-p_i)+p_i z_i = 0$ for at least one i (more than one if z_i has multiplicity greater than 1). From this fact, and the fact that the p_i's are strictly positive, z_i is the real number:

$$z_i = -\frac{1-p_i}{p_i},$$

from which we can solve for

$$p_i = \frac{1}{1 - z_i}$$

to uniquely recover the multiset $[p_1, \ldots, p_n]$ of parameters.

If $f_n = 0$, let $l \leq n$ be the largest integer such that $f_l > 0$ as in the claim. Consider the identity (6) with n replaced by l. The RHS is the degree l polynomial $f_0 + f_1 z + \cdots + f_l z^l$ which implies, following the same reasoning, that we can recover the multiset of l non-zero parameters $[p_1, \ldots, p_l]$. The full set of parameters is thus $\left[p_1, \ldots, p_l, 0^{(n-l)}\right]$. ■

Next, we turn to identification in the multinomial case.

Proof of Theorem 2: Without loss of generality, we take X to be the standard basis $E = \{e_1, \ldots, e_k\}$ of R^k. For an E-valued random variable x_i, let $p_i = (p_{i1}, \ldots, p_{ik})$ denote its probability distribution and write $x^{(j)}$ for the indicator of the event $x_i = e_j$. Then the *multivariate probability generating function* of such x_i is

$$\begin{aligned} G_{x_i}(s_1, \ldots, s_k) &\equiv E s_1^{x_i^{(1)}} s_2^{x_i^{(2)}} \cdots s_k^{x_i^{(k)}} \\ &= p_{i1} s_1 + \cdots + p_{ik} s_k, \end{aligned} \tag{7}$$

where (7) follows from the fact that in any realization, one of the $x_i^{(j)}$'s is 1, and all the others are 0.

Define the statistic $\xi(x_1, \ldots, x_n) = \sum_i x_i \in R^k$ and its generating function:

$$\begin{aligned} G_\xi(s_1, \ldots, s_k) &= E\left[s_1^{x_1^{(1)} + \cdots + x_n^{(1)}} s_2^{x_1^{(2)} + \cdots + x_n^{(2)}} \cdots s_k^{x_1^{(k)} + \cdots + x_n^{(k)}}\right] \\ &= \prod_{i=1}^{n} E\left[s_1^{x_i^{(1)}} s_2^{x_i^{(2)}} \ldots s_k^{x_i^{(k)}}\right] \qquad (8) \\ &= \prod_{i=1}^{n} \Big(\sum_{j=1}^{k} p_{ij}\, s_j\Big) \qquad (9) \end{aligned}$$

where (8) follows from independence, and (9) follows from the fact that the $x_i^{(j)}$'s are 0-1 variables.

Suppose that two multisets $[p_1, \ldots, p_n]$ and $[p'_1, \ldots, p'_n]$ generate the same distribution of ξ. Then the generating function of ξ must be the same under

the two multisets, so

$$\prod_{i=1}^{n}\Big(\sum_{j=1}^{k} p_{ij}\, s_j\Big) \;=\; \prod_{i=1}^{n}\Big(\sum_{j=1}^{k} p'_{ij}\, s_j\Big) \tag{10}$$

is an identity in the s_j's.

Each of the terms $\sum_{j=1}^{k} p_{ij}\, s_j$ and $\sum_{j=1}^{k} p'_{ij}\, s_j$ is an irreducible polynomial of degree 1 in the ring $\mathcal{R}(s_1, \ldots, s_k)$ of k-variable polynomials over the real numbers. Since $\mathcal{R}(s_1, \ldots, s_k)$ is a Unique Factorization Domain, both sides of (10) have the same irreducible factors. That is, there must be a permutation $\varrho : \{1, \ldots, n\} \to \{1, \ldots, n\}$, such that, for every i,

$$\sum_{j=1}^{k} p_{ij}\, s_j \;=\; \sum_{j=1}^{k} p'_{\varrho(i)j}\, s_j.$$

This implies that $p_i = p'_{\rho(i)}$, thus $[p_1, \ldots, p_n] = [p'_1, \ldots, p'_n]$. ■

The p_{ij}'s consist of $n \times (k-1)$ independent parameters, a much smaller number than the $\binom{n+k-1}{k-1} - 1$ parameters needed to define the distribution of ξ.[13] This suggests that a simpler set of statistics that has the "right" number of parameters may suffice. This, however, is not straightforward. For example, a natural family of statistics to consider is the k marginals: $\xi^{(j)} = x_1^{(j)} + \cdots + x_n^{(j)}$. Each statistic in this family has n parameters, for a total of nk parameters. These statistics are not enough to identify the multiset $[\mu_1, \ldots, \mu_n]$, however. Suppose $n = k = 3$ and apply Theorem 1 to each of $\xi^{(j)}, j = 1, 2, 3$, to obtain a multiset of Bernoulli probabilities shown in the column of the following table:

$M(1)$	$M(2)$	$M(3)$
0.3	0.3	0.4
0.4	0.4	0.2
0.2	0.5	0.3

[13] This formula follows from the Stars and Bars Theorem: there are $\binom{n+k-1}{k-1}$ ways to allocate n objects into k non-negative integers such that $m_1 + m_2 + \cdots + m_k = n$. The distribution of ξ therefore involves $\binom{n+k-1}{k-1} - 1$ degrees of freedom. In the Bernoulli case ($k = 2$), this reduces to $\binom{n+2-1}{2-1} - 1 = n$, which is the same as the number of unknown probabilities: $n \times (2-1) = n$. When $k > 2$, on the other hand, the number of parameters defining the distribution of ξ is $\frac{\binom{n+k-1}{k-1}-1}{n(k-1)}$ times, or about $\frac{n^{k-2}}{(k-1)!}$ for large n, larger than the number of parameters that need to be estimated.

These multisets of marginals are consistent with two distinct models:

Model A	Model B
(0.3, 0.3, 0.4)	(0.3, 0.5, 0.2)
(0.4, 0.4, 0.2)	(0.4, 0.3, 0.3)
(0.2, 0.5, 0.3)	(0.2, 0.4, 0.4)

The distributions labeled Model A correspond to the rows of the original table. Model B produces distinct probability distributions, yet it is also consistent with the information contained in the marginals. However, Model A and B produce different distributions for the statistic ξ (for example, the probability of $\xi = (0, 1, 2)$ is 0.106 under Model A but 0.108 under Model B).[14]

Proof of Theorem 3: Denote the Borel σ-algebra of $\mathcal{X}$ by $\tilde{\mathcal{X}}$ and let $\{\mathcal{F}_t\}_{t=1}^{\infty}$ be a filtration, each element of which is the algebra generated by a finite partition of $\mathcal{X}$, and such that the σ-algebra generated by $\cup_{t=1}^{\infty}\mathcal{F}_t$ is $\tilde{\mathcal{X}}$.[15]

By way of contradiction, suppose that there are two distinct multisets $\boldsymbol{\mu}$ and $\hat{\boldsymbol{\mu}}$ such that:

$$\Phi(\boldsymbol{\mu}) = \Phi(\hat{\boldsymbol{\mu}}) = \mathbf{P}. \tag{11}$$

It is convenient to express these multisets as flattenings:

$$\begin{aligned} \boldsymbol{\mu} &= \{(\mu_l, r) : l = 1, \dots L, r = 1, \dots, m_l\}; \\ \hat{\boldsymbol{\mu}} &= \{(\hat{\mu}_{\hat{l}}, \hat{r}) : \hat{l} = 1, \dots \hat{L}, \hat{r} = 1, \dots, \hat{m}_{\hat{l}}\}, \end{aligned} \tag{12}$$

where the μ_l's and $\mu_{\hat{l}}$'s are distinct. By Dynkin's $\lambda - \pi$ Theorem, adapted to our context, there exists t and a partition $\mathcal{A} = \{A_1, \dots, A_k\}$ in $\mathcal{F}_t$ such that, for any distinct pair $\upsilon, \upsilon' \in \{\mu_1, \dots, \mu_L, \hat{\mu}_1, \dots, \hat{\mu}_{\hat{L}}\}$, there is $A \in \mathcal{A}$ such that $\upsilon(A) \neq \upsilon'(A)$.

For $l = 1, \dots, L$, we take the marginal distribiution of μ_l on the elements of the partition $\mathcal{A}$ to define the Poisson Multinomial distribution

$$p_l \equiv (p_{l1}, \dots, p_{lk}) = \big(\mu_l(A_1), \dots, \mu_l(A_k)\big).$$

[14] These are 0.018+0.048+0.04=0.106 and 0.06+ 0.024+ 0.024=0.108.

[15] That such a filtration exists is standard: take $\mathcal{F}_j$ to be the algebra of sets in $[0, 1]$ generated by the disjoint intervals of length j and use the fact that every standard Borel space is isomorphic with a subset of $[0, 1]$ with its Borel σ-algebra.

and the multiset

$$\boldsymbol{p} = \{(p_l, r) : l = 1, \dots L, r = 1, \dots, m_l\}.$$

Note that, for each l, m_l has the same value as in (12). Define the multiset $\hat{\boldsymbol{p}}$ similarly using $\hat{\boldsymbol{\mu}}$.

Define the statistic $\xi = x_1 + \cdots + x_n$. The distribution of ξ under $\boldsymbol{\mu}$ is equal to that under $\boldsymbol{p}$ since this distribution depends only on the probabilities of the partition elements in $\mathcal{A}$ and the multiplicities m_l, both being preserved in the construction of $\boldsymbol{p}$. Similarly, the distribution of ξ under $\hat{\boldsymbol{\mu}}$ is equal to that under $\hat{\boldsymbol{p}}$. Furthermore, (11) implies that ξ has the same distribution under $\boldsymbol{\mu}$ and $\hat{\boldsymbol{\mu}}$ since ξ is a symmetric function. These observations imply that $\boldsymbol{p}$ and $\hat{\boldsymbol{p}}$ generate the same distribution for ξ.

By Theorem 2, the multisets $\boldsymbol{p}$ and $\hat{\boldsymbol{p}}$ are equal. Thus, there is a stabilizer ϱ between them, *i.e.,* a bijection between $\{(p_l, 1), \dots, (p_l, m_l)\}$ and $\{(\hat{p}_{\hat{l}}, 1), \dots, (\hat{p}_{\hat{l}}, \hat{m}_{\hat{l}})\}$ such that $\varrho(p_l, r) = (\hat{p}_{\hat{l}}, \hat{r}) \implies p_l = \hat{p}_{\hat{l}}$ for each $l = 1, \dots L$, ϱ. From this it follows that $m_l = \hat{m}_{\hat{l}}$.

If $\varrho(p_l, r) = (\hat{p}_{\hat{l}}, \hat{r})$ but $\mu_l \neq \hat{\mu}_{\hat{l}}$, then there must be $A_k \in \mathcal{A}$ such that $p_{lk} = \mu_l(A_k) \neq \hat{\mu}_{\hat{l}}(A_k) = \hat{p}_{\hat{l}k}$, contradicting the fact that $p_l = \hat{p}_{\hat{l}}$. This, and the fact that $m_l = \hat{m}_{\hat{l}}$ implies that, for each l, there is a bijection between $\{(\mu_l, 1), \dots, (\mu_l, m_l)\}$ and $\{(\hat{\mu}_{\hat{l}}, 1), \dots, (\hat{\mu}_{\hat{l}}, \hat{m}_{\hat{l}})\}$. A contradiction with the assumption that $\boldsymbol{\mu} = \hat{\boldsymbol{\mu}}$. ■

We note that the proof cannot be reduced to the binomial case—at least not in any simple way. Suppose that $n = 2$ and that, in the last proof, we get a partition into four sets, A_1, A_2, A_3, and A_4. Consider two distinct multisets $\boldsymbol{\mu}$ and $\hat{\boldsymbol{\mu}}$ whose associated multisets of multinomials, $\boldsymbol{\mu}_{|\mathcal{A}} = [(\mu_1(A_j))_{j=1}^4, (\mu_2(A_j))_{j=1}^4]$ and $\hat{\boldsymbol{\mu}}_{|\mathcal{A}} = [(\hat{\mu}_1(A_j))_{j=1}^4, (\hat{\mu}_2(A_j))_{j=1}^4]$, are given by:

$$\begin{gathered} \mu_{1,\cdot} = (1/6, 1/6, 1/3, 1/3), \quad \mu_{2,\cdot} = (1/3, 1/3, 1/6, 1/6) \\ \text{and} \\ \hat{\mu}_{1,\cdot} = (1/6, 1/3, 1/3, 1/6), \quad \hat{\mu}_{2,\cdot} = (1/3, 1/6, 1/6, 1/3). \end{gathered} \tag{13}$$

Pick any of the partition elements $A \in \mathcal{A}$ and its complement. Using the indicator $\mathbf{1}_A(x)$, we can apply Theorem 1 to identify $p_1 = P(x_1 \in A) = 1/3$ and $p_2 = P(x_2 \in A) = 1/6$, where the ordering is arbitrary. The multisets

$[\mu_1(A), \mu_2(A)]$ and $[\hat{\mu}_1(A), \hat{\mu}_2(A)]$ are both equal to $[p_1, p_2] = [1/3, 1/6]$, even though $\boldsymbol{\mu}$ and $\hat{\boldsymbol{\mu}}$ are clearly distinct. The multinomial case dealt with in Theorem 2, on the other hand, would reveal the complete vectors of probabilities in (13).

A.3 Standard Borel Spaces

This section records standard facts from Descriptive Set Theory used throughout the paper. Proofs are provided for completeness. For references, see Kechris (1995, 2023).

A measurable space $(Z, \mathcal{Z})$ is a *standard Borel space* if $\mathcal{Z}$ is the Borel σ-algebra generated by a Polish topology on Z (*i.e.,* a topology that is separable, metrizable, and complete). The set of Borel probability measures on $(Z, \mathcal{Z})$, denoted $\Delta(Z, \mathcal{Z})$, is itself a standard Borel space with Borel σ-algebra $\mathrm{B}(\Delta(Z, \mathcal{Z}))$ (Kechris, 1995, Theorem 17.23).

Here and everywhere else in the paper, $\mathrm{B}(\Delta(Z, \mathcal{Z}))$ is defined as the Borel σ-algebra generated by the topology of weak convergence on $\Delta(Z, \mathcal{Z})$, i.e., $\nu_j \xrightarrow{w} \nu$, iff $\int f\, d\nu_j \to \int f\, d\nu$ for every bounded continuous functions f. Equivalently, $\mathrm{B}(\Delta(Z, \mathcal{Z}))$ is also the σ-algebra generated by the evaluation maps $\nu \xmapsto{e_A} \nu(A)$ for all A in a family of sets $\mathcal{Z}' \subset \mathcal{Z}$ that generates $\mathcal{Z}$ (Kechris, 1995, Theorem 17.24). That is, $\mathcal{Z}$ is the smallest σ-algebra that makes these evaluation maps measurable.

The projection map:

$$Q : (S, \Sigma) \to (S_{/G}, \mathscr{M})$$

defined by:

$$Q(x_1, \ldots, x_n) = [x_1, \ldots, x_n]$$

takes a (labeled) vector $(x_1, \ldots, x_n)$ to the corresponding (unlabeled) multiset $[x_1, \ldots, x_n]$. Recall the equivalence relation $\sim$ on S defined by the finite group of permutations G. The quotient space $(S_{/G}, \mathscr{M})$ of $\sim$-equivalence classes is defined by the quotient σ-algebra

$$\mathscr{M} = \{A \subset S_{/G} : Q^{-1}(A) \in \Sigma\},$$

i.e., the smallest σ-algebra on $S_{/G}$ that makes the projection map Q measurable. The next lemma follows from the fact that the $\sim$-equivalence classes are finite:

Lemma A.1. *$(S_{/G}, \mathscr{M})$ is a standard Borel space.*

Proof: By Proposition 3.12 in Kechris (2023), $(S_{/G}, \mathscr{M})$ is standard Borel if and only if $\sim$ is *smooth*: there is a standard Borel space Y and Borel measurable function $\tau : S \to Y$ such that

$$s \sim s' \iff \tau(s) = \tau(s'). \tag{†}$$

To construct τ, we first note that, since S is standard Borel, there is a Borel isomorphism $f : S \to [0,1]$ into the unit interval (Kechris, 1995, Theorem 15.6). For $s \in S$, define $\tau(s)$ to be the f-minimal element in the orbit of s. Clearly, $\tau : S \to S$ satisfies (†); to show smoothness of $\sim$, we only need to prove that τ is Borel measurable.

To each permutation ϱ corresponds the bijection:

$$(x_1, \ldots, x_n)A \xmapsto{T_\varrho} B(x_{\varrho(1)}, \ldots, x_{\varrho(n)}).$$

In fact, T_ϱ is a Borel automorphism on S since each of its n coordinates, $\mathrm{proj}_j \circ T_\varrho$, is Borel measurable. From this it follows that, for each pair of permutations ϱ, $\varrho^\star$, the set $\{s : f(T_{\varrho^\star}(s)) \leq f(T_\varrho(s))\}$ is Borel, since f, T_ϱ and $T_{\varrho^\star}$ are Borel functions, and $\leq$ is a Borel relation in $[0,1]$.

This implies that for every $\varrho^\star$, the set:

$$B_{\varrho^\star} = \bigcap_{\varrho \in G} \{s : f(T_{\varrho^\star}(s)) \leq f(T_\varrho(s))\}$$

is Borel. Furthermore, the sets $B_1, \ldots, B_{n!}$ form a Borel cover of S, for any fixed but arbitrary enumeration $\varrho_1, \ldots, \varrho_{n!}$ of G. Define the partition

- $A_1 = B_{\varrho_1}$,
- $A_2 = B_{\varrho_2} \backslash A_1$,
- $\ldots$
- $A_{n!} = B_{\varrho_{n!}} \backslash \cup_{j=1}^{n!-1} A_j$.

Finally, define

$$\tau(s) = T_{\varrho_j}(s) \text{ for } s \in A_j.$$

Then τ is a Borel measurable function since T_{ϱ_j} is an automorphism and A_j is a Borel set. Thus, $\sim$ is smooth, and the claim of the lemma follows. ■

While the space $(S, \mathscr{S})$ of outcome profiles with the σ-algebra of symmetric events is not a standard Borel space (since it does not separate points), the space $\Delta(S, \mathscr{S})$ of aggregate distributions is:

Lemma A.2. *$\Delta(S, \mathscr{S})$ is a standard Borel space and the mapping*

$$Q^\star : \Delta(S, \mathscr{S}) \to \Delta(S_{/G}, \mathscr{M}),$$

defined by:

$$Q^\star(\mathbf{P})(A) = \mathbf{P}(Q^{-1}(A)), \text{ for } A \in \mathscr{M}$$

is a Borel isomorphism.

Proof: Since $(S_{/G}, \mathscr{M})$ is standard Borel by Lemma A.1, so is $\Delta(S_{/G}, \mathscr{M})$ with the σ-algebra induced by weak convergence described above. Let $\mathbf{P}_j \to \mathbf{P}$ be a convegent sequence in $\Delta(S, \mathscr{S})$ and let $\nu_j = Q^*(\mathbf{P}_j)$ and $\nu = Q^*(\mathbf{P})$ be the corresponding sequence in $\Delta(S_{/G}, \mathscr{M})$. Since Q is a bijection between $\mathscr{S}$ and $\mathscr{M}$, it is easy to see that $\mathbf{P}_j(Q^{-1}(A)) \to \mathbf{P}(Q^{-1}(A))$, iff $\nu_j(A) \to \nu(A)$ for all $A \in \mathscr{M}$. Thus, weak convergence of ν_j to ν in $(S_{/G}, \mathscr{M})$ is equivalent to weak convergence of $\mathbf{P}_j$ to $\mathbf{P}$. This shows that Q^* preserves the topology and is therefore a Borel isomorphism. ■

Lemma A.3. *The set $\Delta_{\text{Indep}}(S)$ of independent probability distributions is a standard Borel subspace of $\Delta(S) \equiv \Delta(X \times \cdots \times X)$ when equipped with the subspace σ-algebra.*

Proof: Consider the standard Borel space $(\Delta(X))^n$ equipped with the product σ-algebra. Define the mapping

$$\Xi : (\Delta(X))^n \to \Delta(S)$$

by $(\mu_1, \ldots, \mu_n) \mapsto \mu_1 \times \cdots \times \mu_n$. This map is clearly injective. It is also standard to show that it is continuous with respect to the weak convergence topology, and therefore Borel. By the Lusin-Souslin Theorem (Kechris, 1995, Theorem 15.1), $\Xi\Big(\big(\Delta(X)\big)^n\Big)$ is a Borel subset of $\Delta(S)$, and is thus itself a standard Borel space. It follows directly from the definition of Ξ that $\Xi\Big(\big(\Delta(X)\big)^n\Big) = \Delta_{\text{Indep}}(S)$. ■

A.4 Remaining Proofs

Lemma A.4. *The mapping*

$$\Phi : \Delta_{\text{Indep}}(S) \to \Delta(S, \mathscr{S})$$

is Borel measurable and its graph

$$\text{Graph}(\Phi) = \{(P, \mathbf{P}) : P \in \Delta_{\text{Indep}}(S) \text{ and } \Phi(P) = \mathbf{P}\}$$

and range $\Phi(\Delta_{\text{Indep}}(S))$ *are Borel sets.*

Proof: Since the evaluation maps $e_A, A \in \mathscr{S}$ generate the σ-algebra $\text{B}(\Delta(S, \mathscr{S}))$, Φ is Borel measurable provided $P \mapsto e_A \circ \Phi(P)$ is for each symmetric event $A \in \mathscr{S}$. Since for any such event A, $e_A \circ \Phi(P) = P(A)$, this reduces to $P \mapsto P(A)$ which is measurable by the definition of $\text{B}(\Delta_{\text{Indep}}(S))$.

From Kechris (1995, Theorem 14.12), it follows that $\text{Graph}(\Phi)$ is a Borel set. Theorem 3, implies that the section $(\mathbf{P} \times \Delta_{\text{Indep}}(S)) \cap \text{Graph}(\Phi)$ is finite for every $\mathbf{P} \in \Delta(S, \mathscr{S})$. By the Lusin-Novikov Theorem (Kechris, 1995, Theorem 18.10), which states that the projection of a Borel set with countable sections is Borel, $\Phi(\Delta_{\text{Indep}}(S)) = \text{proj}_{\Delta(S,\mathscr{S})} \text{Graph}(\Phi)$ is Borel. ■

The Borel measurability of $\Phi(\Delta_{\text{Indep}}(S))$ in the last lemma requires the identification results established earlier. In general, the projection of $\text{Graph}(\Phi)$ on $\Delta(S, \mathscr{S})$ is only guaranteed to be an analytic set—not necessarily Borel. Theorem 3 establishes that the fibers $\Phi^{-1}(\mathbf{P})$ are finite (possibly empty, if $\mathbf{P}$ is not in the range of Φ), making it possible to apply the Lusin-Novikov Theorem to conclude that $\Phi(\Delta_{\text{Indep}}(S))$ is, in fact, Borel.

Proof of Theorem 4: For a player i, let ν_i denote the marginal distribution of $\tilde{P}_i$ on $\Delta(S, \mathscr{S})$. By Rokhlin's Disintegration Theorem (Rokhlin (1962), Simmons (2012)), there is a ν_i-unique measurable kernel

$$\kappa_i : \Delta(S, \mathscr{S}) \to \Delta(\Delta_{\text{Indep}}(S)),$$

satisfying

1. $\tilde{P}_i(B) = \int (\kappa_i(\mathbf{P}) \times \delta_{\mathbf{P}})(B)\, d\nu_i(\mathbf{P})$ for any measurable $B \subset \Delta_{\text{Indep}}(S) \times \Delta(S, \mathscr{S})$;[16] and

[16] The kernel κ_i is defined for all values of $\mathbf{P}$, including ones outside the range of $\tilde{P}_i$.

2. $\kappa_i(\mathbf{P})(\Phi^{-1}(\mathbf{P})) = 1$, for $\mathbf{P}$ ν_i-a.s.

The kernel κ_i should be interpreted as the regular conditional probability of $\tilde{P}_i$ with respect to the mapping Φ. That is, $\kappa_i(\mathbf{P})(\cdot) = \tilde{P}_i(\cdot \,|\, \mathbf{P})$. From Theorem 3, for every $\mathbf{P}$ in the range of Φ, there is a unique multiset $\boldsymbol{\mu} = [\mu_1, \ldots, \mu_n]$ such that:

$$\Phi^{-1}(\mathbf{P}) = \big\{\mu_{\pi(1)} \times \cdots \times \mu_{\pi(n)} : \pi \text{ is an assignment of } \boldsymbol{\mu}\big\}.$$

Since $\Phi^{-1}(\mathbf{P})$ is finite, $\kappa_i(\mathbf{P})(\Phi^{-1}(\mathbf{P}))$ has the desired form:

$$\sum_{\pi} \big(\mu_{\pi(1)} \times \cdots \times \mu_{\pi(n)}\big) \; w_i(\pi),$$

where the w_i's are non-negative and $\sum_\pi w_i(\pi) = 1$, with the sum ranging over the finite set of assignments of $\boldsymbol{\mu}$.

For any $A \subset \Delta(S, \mathscr{S})$ and player $\bar{\imath}$, $\nu_{\bar{\imath}}(A) = 1$ means $\tilde{P}_{\bar{\imath}}\,(\Delta_{\text{Indep}}(S) \times A) = 1$, by definition. Since the $\tilde{P}_i$'s are mutually absolutely continuous, it follows that $\nu_i(A) = 1$ for all i. The kernel κ_i has the properties above on every set A with ν_i-probability 1. By the earlier observation, this means we can find a single set A with ν_i-probability 1 for all i, as required. ■

Proof of Theorem 5: The first part of the Assumption 3 implies

$$\sum_{s=1}^{\infty} \frac{1}{s^2} \operatorname{Var}[f(z^s)] < \infty.$$

Thus, the strong law of large numbers applies, and therefore,

$$F_t - E[F_t] \longrightarrow 0, \;\; Q\text{-a.s.}$$

Since $E[F_t] \leq 0$ under $\mathbf{H}_0$ per the second part of Assumption 3, we have

$$\limsup_{t\to\infty} F_t \leq \limsup_{t\to\infty} (F_t - E[F_t]) = 0, \;\; Q\text{-a.s.}$$

under $\mathbf{H}_0$, and the first claim follows.

For the second part, split the sample into $\mathcal{S}_{A,t} = \{s \leq t \mid (\alpha_1^s, \ldots, \alpha_n^s) \in \mathcal{A}\}$ and $\mathcal{S}_{N,t} = \{s \leq t \mid (\alpha_1^s, \ldots, \alpha_n^s) \notin \mathcal{A}\}$. Construct the two partial sums

$$F_{t,A} = \frac{1}{t} \sum_{s \in \mathcal{S}_{A,t}} f(z^s)$$

$$F_{t,N} = \frac{1}{t} \sum_{s \in \mathcal{S}_{N,t}} f(z^s)$$

and observe that $F_t = F_{t,A} + F_{t,N}$. With the first part of Assumption 3, we have

$$F_{t,A} - E[F_{t,A}] \longrightarrow 0, \ Q\text{-a.s.}$$

either due to $\Psi_t \to 0$ or due to the strong law of large numbers. Note that $F_{t,N} \leq (1 - \Psi_t)M$ with the first part of Assumption 3. With the second part of Assumption 3, we have

$$\begin{aligned} 1 - \Psi_t &\geq \frac{1}{M} F_{t,N} \\ &\geq \frac{1}{M} \left(F_{t,N} + E[F_{t,A}]\right) \end{aligned}$$

and, hence, Q-a.s.:

$$\begin{aligned} \liminf_{t\to\infty}(1 - \Psi_t) &\geq \frac{1}{M} \liminf_{t\to\infty} \left(F_{t,N} + E[F_{t,A}]\right) \\ &= \frac{1}{M} \liminf_{t\to\infty} \left(F_{t,N} + E[F_{t,A}]\right) \\ &\quad + \frac{1}{M} \lim_{t\to\infty} \left(F_{t,A} - E[F_{t,A}]\right) \\ &= \frac{1}{M} \liminf_{t\to\infty} \left(F_{t,N} + F_{t,A}\right) \\ &= \frac{1}{M} \liminf_{t\to\infty} F_t > 0 \end{aligned}$$

or,

$$\limsup_{t\to\infty} \Psi_t < 1.$$

■

To highlight the role of part 2 of Assumption 3, suppose that $n = 2$, $x_i^t \in \{0, 1\}$, and the action space is $A = \{e, o\}$ (for "even" and "odd"). The players do not observe the realizations of the x_i^t's, but agree that they are i.i.d. with $\Pr(x_i^t) = 0$ for all i and t.

Consider the coordination game where each player's payoff depends on the realized value of $x_1^t + x_2^t$ as follows: the payoff is 1 if both correctly predict whether $x_1^t + x_2^t$ is even or odd, -1 if they make the same but incorrect prediction, and -10 if they disagree. There are two pure-strategy equilibria (o, o) and (e, e) in the set of agreement equilibria $\mathcal{A}$, both of which yield an expected payoff of zero. For the statistic $f(z) = v_1 + v_2$, we therefore have

$$Ef(z \,|\, (e, e)) = Ef(z \,|\, (o, o)) = 0.$$

Generate a sequence of draws (x_1^t, x_2^t) and select the equilibrium (o, o) with a fixed probability $p > 0.5$ when $x_1^t + x_2^t$ is odd, and (e, e) with probability p when $x_1^t + x_2^t$ is even. Although the resulting states z^t are i.i.d., the *unconditional* expectation $Ef(z^t)$ is strictly positive in every period, so the conclusion of the theorem no longer holds. For this reason, the (possibly mixed) equilibrium $(\alpha_1^t, \ldots, \alpha_n^t)$ must be included in the state z^t, and the second part of Assumption 3 is needed.

Theorem 7. *To every reduced-form game $G = G(P_1, \ldots, P_n)$ corresponds a unique Bayesian game with* common knowledge of biases $B = B(P_1, \ldots, P_n)$ *defined by the requirement that the belief profile $P_1, \ldots, P_n$ is common knowledge. The games G and B have the same set of equilibrium distributions $\mathcal{N}(P_1, \ldots, P_n)$.*

Proof of Theorem 6: Fix a game $G(P_1, \ldots, P_n)$ and its set of Nash equilibria $\mathcal{N}(P_1, \ldots, P_n)$. We construct the Bayesian game B following Brandenburger and Dekel (1987). The space of uncertainty is the standard Borel space $S = X^n$. Player i's first-order belief is $P_i \in \Delta(S)$. Higher-order beliefs are determined by the assumption that the profile of first-order beliefs $P_1, \ldots, P_n$ is common knowledge. Thus, every player i has only one type $\bar{t}_i$ characterized by $P_i \times \delta_{\bar{t}_{-i}} \in \Delta(S \times T_{-i})$ in which this player puts unit mass on the other types being $\bar{t}_{-i}$.

An equilibrium for B specifies the strategy $\bar{\alpha}_i \in \Delta(A)$ for each player i (since each player has only one type) such that:

$$\bar{\alpha}_i \in \arg\max_{\alpha_i} \mathcal{V}_i(\alpha_i; \bar{\alpha}_{-i}; P_i).$$

This is the same as the definition of equilibrium in $G(P_1, \ldots, P_n)$ in (1), so the equilibria coincide. ■

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B Online Appendix

This section contains additional details and calculations for the introductory example of Section 2.

B.1 The stage game

If only one firm is present and has marginal costs c, it will select a quantity q maximizing its monopolistic profits, $\pi(q) = q(2 - q - c)$ (before deducting entry costs). The solution is $q = 1 - c/2$, $P = 1 + c/2$ and $\pi = (1 - \frac{c}{2})^2$. Thus, $\pi = 1$, if $c = 0$ and $\pi = 1/4$, if $c = 1$. If both firms enter, they compete on quantity in a duopoly. Firm 1 chooses its profit-maximizing quantity q_1, given the quantity q_2 produced by firm 2 and vice versa, where the market price is $P = 2 - q_1 - q_2$. The solution is $q_1 = (2 + c_2 - 2c_1)/3$ and $q_2 = (2 + c_1 - 2c_2)/3$ for the quantities, $P = (2 + c_1 + c_2)/3$ for the market price and $\pi_1 = (2 + c_2 - 2c_1)^2/9$ as well as $\pi_2 = (2 + c_1 - 2c_2)^2/9$ for the profits (before deducting entry costs). Table 1 provides the four combinations of marginal costs and profits that can arise if both firms enter.

firm 1	firm 2	firm 1	firm 2
$c_1 = 0$	$c_2 = 0$	$c_1 = 0$	$c_2 = 1$
$\pi_1 = 4/9$	$\pi_2 = 4/9$	$\pi_1 = 1$	$\pi_2 = 0$
$c_1 = 1$	$c_2 = 0$	$c_1 = 1$	$c_2 = 1$
$\pi_1 = 0$	$\pi_2 = 1$	$\pi_1 = 1/9$	$\pi_2 = 1/9$

Table 1: *Combinations of marginal costs and resulting profits.*

Suppose that there are always two types of firms, a "good" firm and a "bad" firm. For simplicity, we restrict attention to the case where $\mu_g + \mu_b = 1$. The good firm draws $c = 0$ with probability $\mu_g \in [0.5, 1]$, while the bad firm draws $c = 0$ with probability $\mu_b = 1 - \mu_g$. Consider a firm of either type μ. If only that firm enters, its expected monopoly profits are

$$\pi_M(\mu) = \mu + (1 - \mu)\frac{1}{4} = \frac{1 + 3\mu}{4}.$$

If both firms enter, the duopoly profits of that firm are

$$\pi_D(\mu) = \mu(1-\mu)\frac{4}{9} + \mu^2 + (1-\mu)\mu\frac{1}{9} = \mu - \frac{4}{9}\mu(1-\mu).$$

B.2 The entry game

Consider firm i and let a_j be the entry strategy and therefore the entry probability of firm $j = 3 - i$. Suppose that firm i is believed to be a good firm with probability p_i and a bad firm with probability $1 - p_i$. Let

$$\mu_i^e = \mu^e(p_i) = p_i\mu_g + (1-p_i)(1-\mu_g)$$

be the resulting probability for drawing $c = 0$. Across these two possibilities, expected monopoly profits are

$$\pi_{M,i}^e = p_i\pi_M(\mu_g) + (1-p_i)\pi_M(1-\mu_g) = \frac{1+3\mu_i^e}{4},$$

while expected duopoly profits are

$$\pi_{D,i}^e = p_i\pi_D(\mu_g) + (1-p_i)\pi_D(1-\mu_g) = \mu_i^e - \frac{4}{9}\mu_g(1-\mu_g)$$

since $\mu_b = 1 - \mu_g$ and thus $\mu_b(1-\mu_b) = \mu_g(1-\mu_g)$. Firm i will enter if its entry costs κ_i are below the threshold a_i, which is equal to expected profits:

$$\begin{aligned}
\kappa_i &\leq a_i = r(a_j; p_i) \\
&= (1-a_j)\pi_{M,i}^e + a_j\pi_{D,i}^e \\
&= \frac{1+3\mu_i^e}{4} - \left(\frac{1}{4}(1-\mu_i^e) + \frac{4}{9}\mu_g(1-\mu_g)\right)a_j.
\end{aligned}$$

In an agreement equilibrium, we have $p_2 = 1-p_1$. Note that $\mu_2^e = \mu^e(1-p) = 1 - \mu_1^e$. We therefore have to solve

$$\begin{aligned}
a_1 &= r(a_2, p_1) = \frac{1+3\mu_1^e}{4} - \left(\frac{1}{4}(1-\mu_1^e) + \frac{4}{9}\mu_g(1-\mu_g)\right)a_2 \\
a_2 &= r(a_1, 1-p_1) = \frac{4-3\mu_1^e}{4} - \left(\frac{1}{4}\mu_1^e + \frac{4}{9}\mu_g(1-\mu_g)\right)a_1
\end{aligned}$$

for a_1 and a_2. The solution is

$$a_1^{\mathrm{A}}(p_1) = \frac{\frac{1+3\mu_1^e}{4} - \left(\frac{1}{4}(1-\mu_1^e) + \frac{4}{9}\mu_g(1-\mu_g)\right)\frac{4-3\mu_1^e}{4}}{1 - \left(\frac{1}{4}(1-\mu_1^e) + \frac{4}{9}\mu_g(1-\mu_g)\right)\left(\frac{1}{4}\mu_1^e + \frac{4}{9}\mu_g(1-\mu_g)\right)}$$

and $a_2 = a_1^{\mathrm{A}}(1-p_1)$, i.e., replacing μ_1^e with $\mu_2^e = 1-\mu_1^e$. In the symmetric rational expectations equilibrium, $p_1 = p_2 = 1/2$ and therefore $\mu^e = 1-\mu^e = 1/2$ and $a_1 = a_2 = a^{\mathrm{RE}}$ with

$$a^{\mathrm{RE}} = \frac{5}{9 + \frac{32}{9}\mu_g(1-\mu_g)}.$$

In the symmetric rational disagreement equilibrium,

$$a_1 = r(a_1; p_1) = \frac{1+3\mu_1^e}{4} - \left(\frac{1}{4}(1-\mu_1^e) + \frac{4}{9}\mu_g(1-\mu_g)\right) a_1$$

The solution is $a_1 = a_2 = a^{\mathrm{D}}(p)$, with $p = p_1$, where

$$a^{\mathrm{D}}(p) = \frac{1+3\mu_1^e}{5-\mu_1^e + \frac{16}{9}\mu_g(1-\mu_g)}.$$

Note that $a^{\mathrm{D}}(0.5) = a^{\mathrm{RE}}$. An asymmetric rational disagreement equilibrium can likewise be calculated by solving the two equations $a_1 = r(a_2, p_1)$ and $a_2 = r(a_1, p_2)$. We skip the tedious details.

B.3 Aggregate statistics and individual probabilities

Denote the marginal cost of the good firm with c and the marginal cost of the bad firm with $\tilde{c}$. In the aggregate, one just sees two cost realizations, without attribution to a good or a bad firm. Recall the relationship of the probabilities $f_0 = \mathrm{Prob}(c+\tilde{c}=0) = \mathrm{Prob}((c,\tilde{c}) = (0,0))$, $f_1 = \mathrm{Prob}((c,\tilde{c}) \in \{(0,1),(1,0)\})$, $f_2 = \mathrm{Prob}((c,\tilde{c}) = (1,1))$ for the three possible oligopoly cost combinations, to the probabilities μ_g and μ_b of drawing $c = 0$ and $\tilde{c} = 0$,

$$f_0 = \mu_g\mu_b,\ f_1 = \mu_g(1-\mu_b) + \mu_b(1-\mu_g),\ f_2 = (1-\mu_g)(1-\mu_b).$$

Since $f_0 + f_1 + f_2 = 1$, these are two independent equations in the two unknowns μ_g and μ_b. Solve the expression of f_1 for μ_b, then replace it in the expression for f_0. This leads to a quadratic equation in μ_g. Since $\mu_g \geq \mu_b$ by assumption, the unique solution is

$$\mu_g = \frac{1}{2}\left(2f_0 + f_1 + \sqrt{4(f_0^2 + f_0 f_1 - f_0) + f_1^2}\right) \text{ and } \mu_b = 2f_0 + f_1 - \mu_g.$$

B.4 Testing

To simplify the exposition, we assume that the same equilibrium is played in all periods. This assumption is not needed in the general testing framework of Section 4.4 or in Theorem 5.

Suppose the game is played t times, $t \to \infty$ with entry and production costs realized i.i.d.. Suppose the same strategies are always chosen, i.e., we are either always in the agreement equilibrium for some fixed $p_1 = p$ and with the probabilities assumed by the agents to be the same as the true data-generating mechanism, or the disagreement equilibrium for some fixed p. An outside observer can calculate the total payoffs $V_t = V_{1,s} + V_{2,s}$ for period $s \in \{1, \ldots, t\}$, i.e., the total profits net of entry costs in period s.

Consider first repeated play of the agreement equilibrium for some fixed $p_1 = p$. Recall that expected profits are zero if $\kappa_1 = a_1^{\mathrm{A}}(p)$ for firm 1. Thus, firm 1 will make $a_1^{\mathrm{A}}(p)/2$ profits net of entry costs on average, conditional on entering, since κ_1 is uniformly distributed on $[0, 1]$. The probability of entering is a_1^{A}. With a similar expression for firm 2, the average total payoffs $\bar{V}_t = (\sum_{s=1}^{t} V_s/t)$ converge to

$$\bar{V}^{\mathrm{A}}(p) = \left(\left(a_1^{\mathrm{A}}(p)\right)^2 + \left(a_1^{\mathrm{A}}(1-p)\right)^2 \right)/2,$$

almost surely.

For the symmetric disagreement equilibrium, and with the duopoly profits π_D per equation (B.1) and conditional on entry, average profits net of average entry costs $a^{\mathrm{D}}(p)/2$ are

$$\begin{aligned} \bar{\pi}^{\mathrm{D}}(p) &= (1 - a^{\mathrm{D}}(p))\frac{5}{8} + \frac{a^{\mathrm{D}}(p)}{2}\left(\pi_D(\mu_g) + \pi_D(1-\mu_g)\right) - \frac{a^{\mathrm{D}}(p)}{2} \\ &= \frac{5}{8} - \left(\frac{5}{8} + \frac{4}{9}\mu_g(1-\mu_g)\right) a_D(p). \end{aligned}$$

Multiplying this by the probability $a^{\mathrm{D}}(p)$ of entry and multiplying it by two (for two firms) delivers average total payouts as

$$\begin{aligned} \bar{V}^{\mathrm{D}}(p) &= 2a^{\mathrm{D}}(p)\bar{\pi}^{\mathrm{D}}(p) \\ &= \frac{5}{4}a^{\mathrm{D}}(p) - \left(\frac{5}{4} + \frac{8}{9}\mu_g(1-\mu_g)\right) a_D(p)^2 \end{aligned}$$

Figure 2 compares the average payoffs for these two equilibria as a function of p. The average disagreement profits are always below this amount, while

the average agreement payoffs are always above this amount, indicated by the black dashed line. This can be used to test whether average profits are outside the agreement range, but still within the disagreement range.